Social Impact Theory: Formalizing a Theory of Social Loafing

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The Objective

The overall goal was to formalize Latané's (1973, as cited in Latané et al., 1979) social impact theory with some extensions.

Verbal Introduction to the Theory

Latané's (1973, as cited in Latané et al., 1979; Latané, 1981) social impact theory distinguishes between people seen as either sources or targets of social impact. The number, their strength and immediacy play a crucial role in the amount of impact experienced. Furthermore, it postulates that there is a division of impact among those targets, which leads to lessened efforts the larger the group is, and the marginal effect of the *n*th person is smaller than that of the (*n*-1)th (for both sources and cotargets) (Latané, 1981; Latané et al., 1979).

Core Constructs of the Theory

There are a few constructs that need to be specified (Figure 1). Due to a lot of jingle-jangle fallacies, we had to compromise. The constructs of impact, pressure, contingency, incentive, and effort were barely defined in the text.

In this model, cotargets (CT) are the group members without one self (important for formalization) who are targets of social impact. The prefix "co" was used in Karau & Williams (1993), but Latané et al. (1979) differentiated between oneself and "other persons also in the target group" (p. 830) as well. The sources (S) exert this social impact. "Source" was adopted from Karau & Williams (1993), which describes "pressures to work come from outside the group" (Latané et al., 1979, p. 830) and "the impact that the experimenters have" (Latané et al., 1979, p. 825) well.

We integrated social pressure and social impact into one construct as P. Both were only mentioned in the context of functions. It was stated, that more S increases P and more CT divides it (Latané et al., 1979).

CON is the "felt […] contingency between their input and the outcome" (Latané et al., 1979, p. 830) and we interpreted it as a prior expectation of whether one's reward is in line with the effort. INC is the "individual's incentive to perform" (Latané et al., 1979, p. 830) which we defined as the "incentive to give full effort" for mathematical formalization.

The construct "effort" as well as "output" could be (often) interpreted as measurable in the text. But sometimes, it was not that clear for "effort", and we also merged our two paths first within the psyche, which is why we defined individual effort (IE) as a psychological and individual outcome (IO) as a measurable variable (Latané et al., 1979).

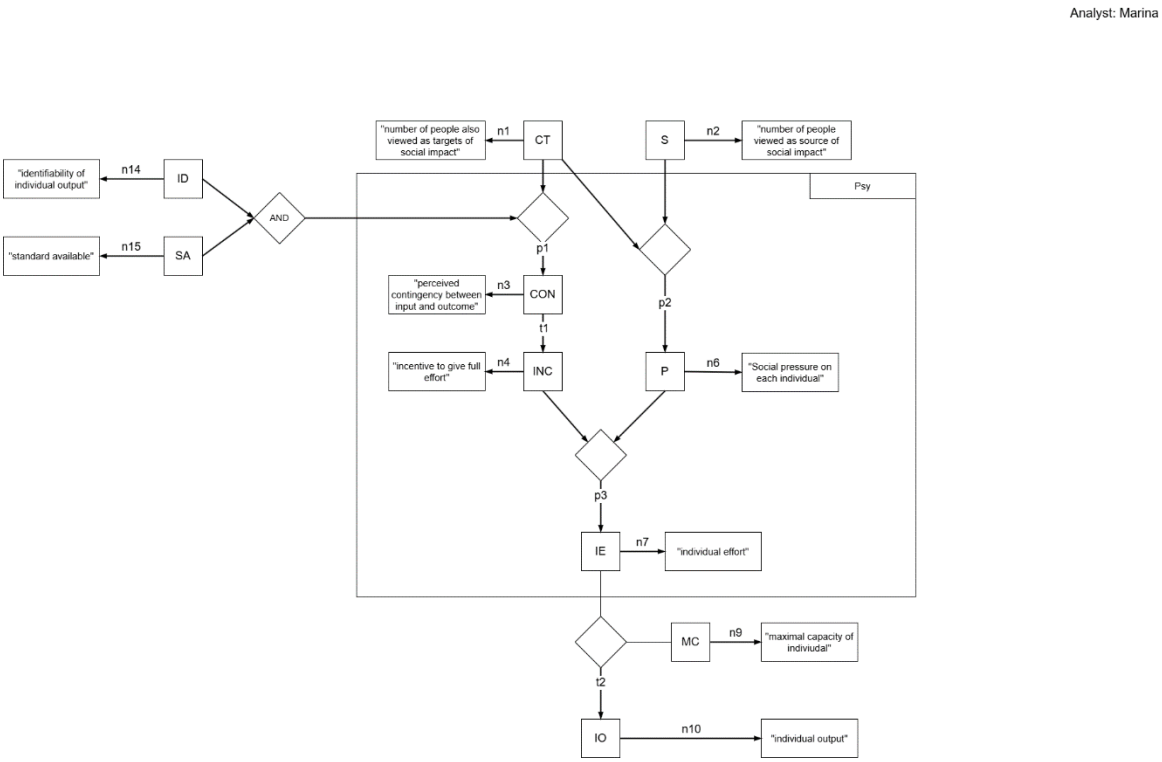
The individual maximum capacity (MC) was needed for the transformation in t1 and represents the maximal output an individual can produce (Latané et al., 1979).

ID and SA are part of the evaluation potential, where ID is to disclose individual outputs within the context of groups, and SA are the standards for comparison (classified into personal, social, and objective; e.g., output of other participants) (Harkins & Jackson, 1985; Karau & Williams, 1993).

See Appendix A for the Construct Table, which shows some relevant quotations.

Figure 1

The mechanism



Note. This visual overview shows parts of the social impact theory, an extension with CON and INC, which can be described in terms of this theory, and, with ID and SA, an assumption of the elimination of social loafing. "Psy" represents the ongoing mechanisms inside the individual (Harkins & Jackson, 1985; Karau & Williams, 1993; Latané et al., 1979).

Relevant Phenomenon

The phenomenon we chose to explain, "social loafing", is the observation of reduced efforts (included in Psy) when individuals work collectively (with CT), compared to when working individually (Karau & Williams, 1993; Latané et al., 1979). Therefore, e.g., the total sound output in Latané et al. (1979) did not equal the sum of IO.

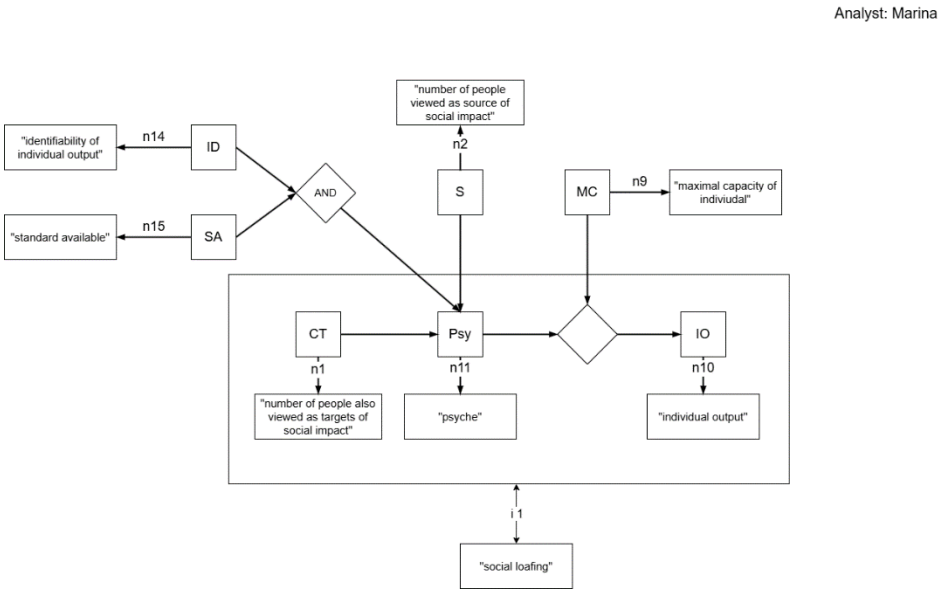
Moderators and coordination loss can be relevant, but they are not part of the main phenomenon (Karau & Williams, 1993; Latané et al., 1979). We decided to display S in Figure 2 due to

Latané et al.'s (1979) emphasis on their importance, even though they were not varied in the original paper.

For a visual overview of the phenomenon, see Figure 2. MC was for operationalization, and Harkins & Jackson (1985) suggested that ID and SA would eliminate social loafing (Karau & Williams, 1993).

Figure 2

The phenomenon



Note. The phenomenon of social loafing (i.e., more CT lead to less IO via Psy). More S can lead to more IO (Karau & Williams, 1993; Latané et al., 1979)). "Psy" is a higher-order concept and represents the ongoing mechanisms inside the individual. MC is primarily for operationalization purposes, and ID and SA were hypothesized to eliminate social loafing (Harkins & Jackson, 1985).

Strength of the evidence: Karau & Williams (1993) included 78 studies with 166 or 163 (they did not explain this discrepancy) effect sizes in total.

Due to variations, a random-effects meta-analysis would have been appropriate (Open Educational Resources des Lehrstuhls Psychologische Methodenlehre und Diagnostik (OERLPMD), 2025b). They did not specify which model they applied, but they weighted the effect sizes by the reciprocal of its variance without mentioning the between-study variances, which indicates the usage of a fixed-effect model (Karau & Williams, 1993; Maitra, 2025). They also calculated a homogeneity statistic and only reported the removal of some studies - without stating the reason for exclusion for each study (Karau & Williams, 1993) (see Appendix D).

Lastly, they did not report p-values or power (OERLPMD, 2025b), but 79% of the included studies implied an effect in the same direction, and the confidence interval excluded zero. The mean weighted *d*s of 0.44 (with outliers) and 0.24 (excluding them) can be labeled as small according to Cohen (Karau & Williams, 1993; OERLPMD, 2026).

Regarding the UTOS dimensions, the following can be stated:

Units (U): Social loafing was tested on and robust for different age groups (from “third grade students or younger” to “organizational employees”), genders, and cultures (Karau & Williams, 1993).

Treatment (T): The phenomenon was verified for different types of tasks, which also differed in their complexity. They made use of physical, cognitive, perceptual, and evaluative tasks as well as maximizing vs. optimizing and additive vs. compensatory vs. disjunctive. Group sizes and moderating variables were also varied (Karau & Williams, 1993).

Outcomes (O): Overall, effect sizes were calculated based on the difference of the effort in an individually (or coactively) vs. a collectively working condition. Because of the different types of tasks, there was the necessity of different measurements (Karau & Williams, 1993).

Settings (S): Social loafing was found in laboratory and field studies, conducted in different countries (Karau & Williams, 1993).

Taking everything into account, the phenomenon of social loafing can be considered - partly cautious - robust because of the variety regarding the UTOS dimensions (Karau & Williams, 1993; Latané et al., 1979).

Formulation of the (Proto-)Theory

Supplementary materials can be found here: https://github.com/marinal304/Social_Loafing.git

The code was written with R (4.3.1) in RStudio (2025.9.1.401).

Latané et al. (1979) never asserted that the social impact theory would fully explain social loafing. Additionally, Karau & Williams (1993) mentioned that the social impact theory has been criticized because it does not address the underlying psychological processes and named several theories for social loafing, but each of them does not consider the numerous variables moderating this phenomenon. However, there is also supporting evidence for the social impact theory (Latané, 1981), and Latané et al. (1979) discussed explanations for the occurrence of social loafing (including CON and INC), which can be described in terms of this theory, and stated it “provides a good account of both the existence and the magnitude of social loafing” (Latané et al., 1979, p. 831).

For the quotations see Appendix B. Within the scope of Latané et al. (1979), especially CT are designated important for the occurrence of social loafing. CT have an effect, first, via P on IE and, thus, on IO (explained by Z1, Z16, Z7, and Z17 combined), and second, via CON and INC (Z4) (Figure 1). CON and INC are mentioned as a possible explanation for social loafing. The (functional) relationship between CON and INC was, however, not really specified. The effect from ID to CON can be inferred from Z4, whereby SA was also integrated because of Z22 and Z23, and from S to P from Z11 (Figure 1) (Harkins & Jackson, 1985; Karau & Williams, 1993; Latané et al., 1979). For more details see Development of a Formal Model.

Development of a Formal Model

Our scope:

First, we excluded coordination loss (e.g., sound cancellation) which is possible through the study design (Latané et al., 1979).

Second, we did include sources because they are an important part of the social impact theory, but we did not vary them in our simulated study. However, they are indirectly component of this simulation due to the beta distribution for MC. Latané et al. (1979) did not report how they measured MC even though they stated S as influential on IE and IO and only used two S.

Third, although strength and immediacy are often stated as relevant (Latané et al., 1979), we excluded them on account of the reduction. These two factors have also been neglected (Karau & Williams, 1993) or rather the focus was on the number dimension (Latané, 1981).

And last, as my extension, ID and SA were included. Due to Latané et al.'s (1979) suggestion that identifiability and rewards contingent are prerequisites for this mechanism, but they actually only talk about "felt" (Latané et al., 1979, S. 830) and "perceived" (Karau & Williams, 1993, S. 684) contingency, which is already represented in the Visual Argument Structure Tool (VAST; Leising et al., 2023) in Figure 1, it was here restricted to identifiability and standards.

The Variable Table can be taken from Appendix A and quotations (e.g., Z1) from Appendix B.

p1: CON = ID*SA + (1-ID*SA) * (1/(CT+1))

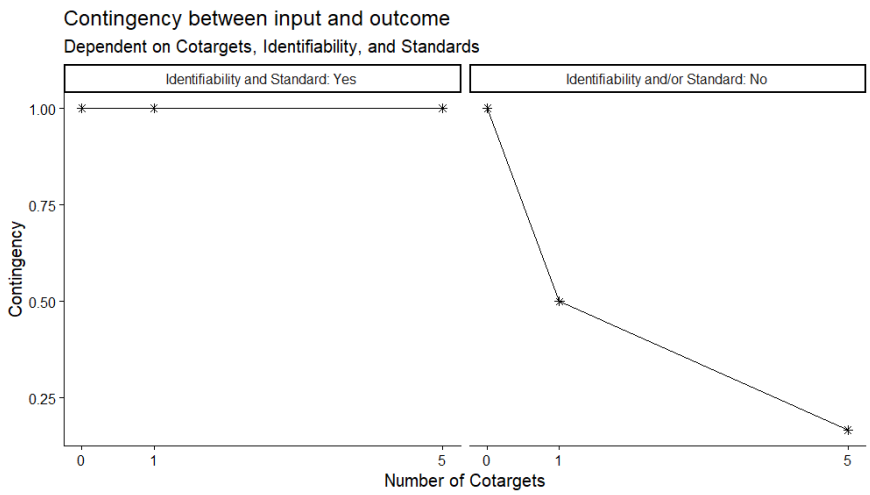
Regarding (1/(CT+1)), this decreasing function is plausible because Latané et al. (1979) assumed a reduced CON in group work (see Z1). Due to Z2, we defined CON to be 1 at zero CT, or when identifiability is given, as individual scores can then be evaluated, too. Since it is argued that SA is crucial for this evaluation (Z22, Z23), and identifiability covered more than just identifiability in prior research, the model and functional relationship were extended by SA in the same manner.

To anticipate t1: The functional relationship between CON and INC was not further specified, and they used those two terms alternately in those paragraphs (Latané et al., 1979). So, we decided on CON = INC (see t1). For this reason, we were also guided by the function 1/n stated in the paper for incentive (see Z3). Therefore, Z4 can be interpreted as affirmative for the result of 1 of p1 if ID and SA (see scope regarding contingent rewards) are given. This is also consistent with Z5.

If ID and SA are given, CON is, independent of CT, 1 (left panel). If ID and/or SA are not given, CON decreases with increasing CT, whereby the *n*th CT has less effect than the (1-*n*)th one (right panel) (Figure 3).

Figure 3

p1-function



p2: P = S/(S + CT + 1)

This function integrates the assumptions in Latané et al. (1979) that both CT and S play a crucial role, whereby we focused only on the number dimension (Z6, Z11). There is a division of P (Z6, Z9, Z10), so that with increasing number of CT, P is decreasing (Z7, Z8, Z9). Additionally, the more S, the more P is felt by each target, which is why P is for 1 S and 0 CT not at its maximum (Z11, Z12). It is also given by p2 that an additional CT or S (*n*th person) has less effect than the (*n*-1)th person (Z12, Z13). Overall, these basic assumptions are in line with Latané (1981).

The individual feels no P in case there are 0 S present (Z6, Z10), and we defined the value range for P as being between 0 and 1, which is also met by p2.

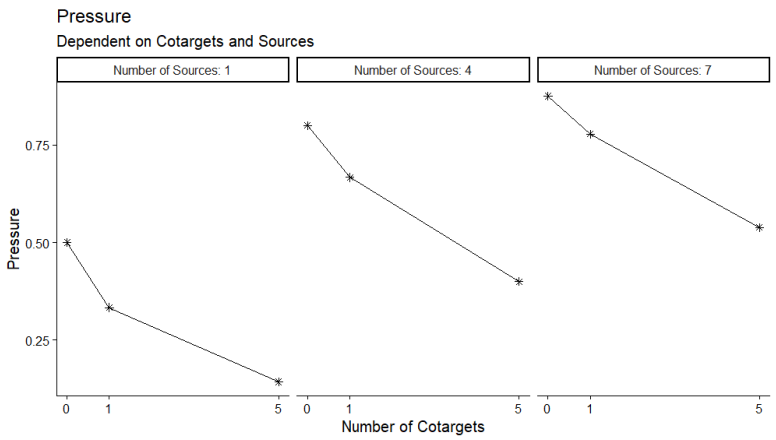
We also incorporated the exponent smaller than one (Z13), whereby the numerator was necessary to represent the influence of S right. Latané et al. (1979) also wrote about an “absolute

value less than one" (Z14), which would contradict our function. But, e.g., they did not integrate CON and INC in the mechanism the same way we did, (1/CT+1) is very well justified and we had to include S, and they reported about the reactions to social stimuli and not direct pressure (Z14).

P drops with increasing CT and grows with increasing S, whereby the *n*th CT and/or S has less effect than the (1-*n*)th one (Figure 4).

Figure 4

p2-function



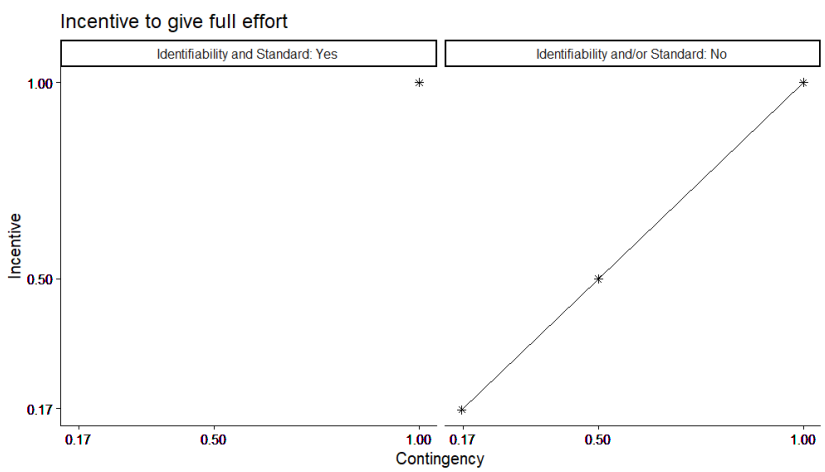
t1: INC = CON

See arguments above for p1.

INC = CON. If ID and SA are given, CON and therefore INC are always 1 (left panel). If ID and/or SA are not given, CON and INC can vary between 0 and 1 (right panel) (Figure 5).

Figure 5

t1-function



p3: IE = 0.5 * INC + 0.5 * P

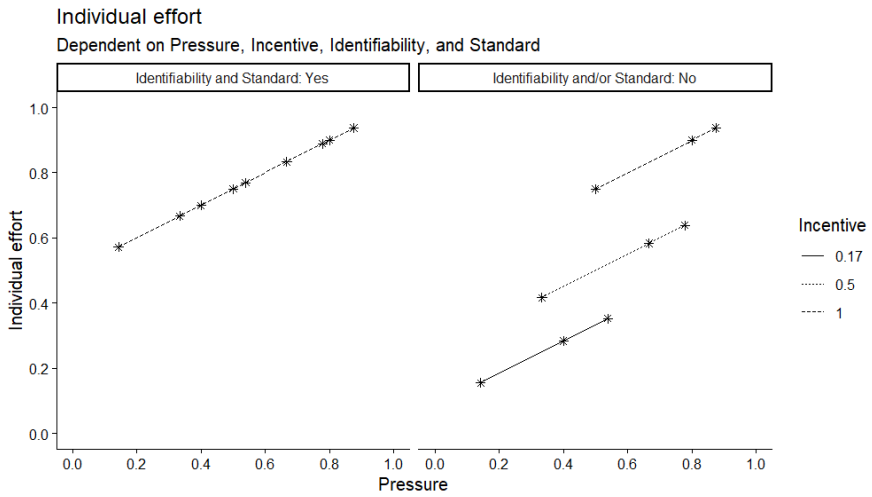
Latané et al. (1979) did not define this weighting of INC and P due to the fact that the mechanism on the left path (Figure 1) was not really correlated with P.

From Z15, which is stated in the context of the argumentation on CON and INC, one can derive that, if ID is not given (and/or SA, see above), an individual may feel less CON and thus less INC with increasing group size, and therefore their IE (and finally IO) is lower. Z16, Z17, Z18, Z19, and Z20 assume that (mediated via P) more CT reduces the individual's IE and IO. Z21 was already explained for p2.

IE grows with both increasing INC and increasing P. If ID and SA are given, INC is always 1, which is why IE is sort of only dependent on P (left panel). If ID and/or SA are not given, INC and P both vary (right panel) (Figure 6).

Figure 6

p3-function



t2: IO = MC * IE

This represents only a transformation from our psychological variable (IE) to a measurable IO and is rather based on common sense. MC is relevant due to the fact that, e.g., someone with damaged vocal cords perhaps has a reduced IO but may have the same level of IE.

IO grows with both increasing MC and increasing IE. If ID and SA are given, IE is always at least 0.5 because CON and therefore INC are always 1 (left panel). If ID and/or SA are not given, IE can vary between 0 and 1 (right panel) (Figure 7).

Because we integrated INC and P for IE, the last part of the last sentence of Z15 (which can refer to IO) may not apply (see also p3).

The superfunction for IE with all measured/manipulated variables, which is integrated in the superfunction for IO, which, in turn, was used for the simulation, also includes random noise because people may differ in, e.g., personality or tiredness.

IO drops with increasing CT and grows with increasing S, whereby the *n*th CT and/or S has less effect than the (1-*n*)th one. If ID and SA are given, CON is always 1, and therefore IO is higher via INC and IE. If ID and/or SA are not given, IO is usually lower. It does not include random noise (see Figure 8 for the superfunction for IO without the random noise).

Figure 7

t2-function

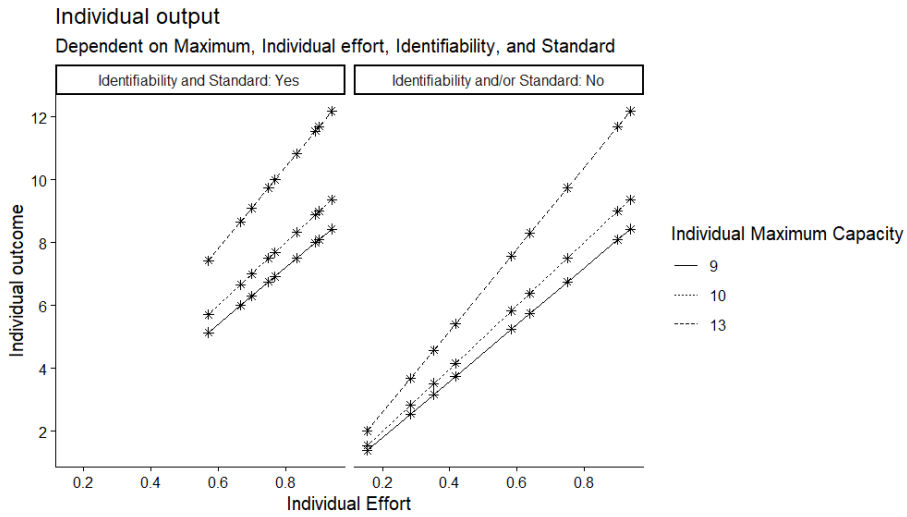
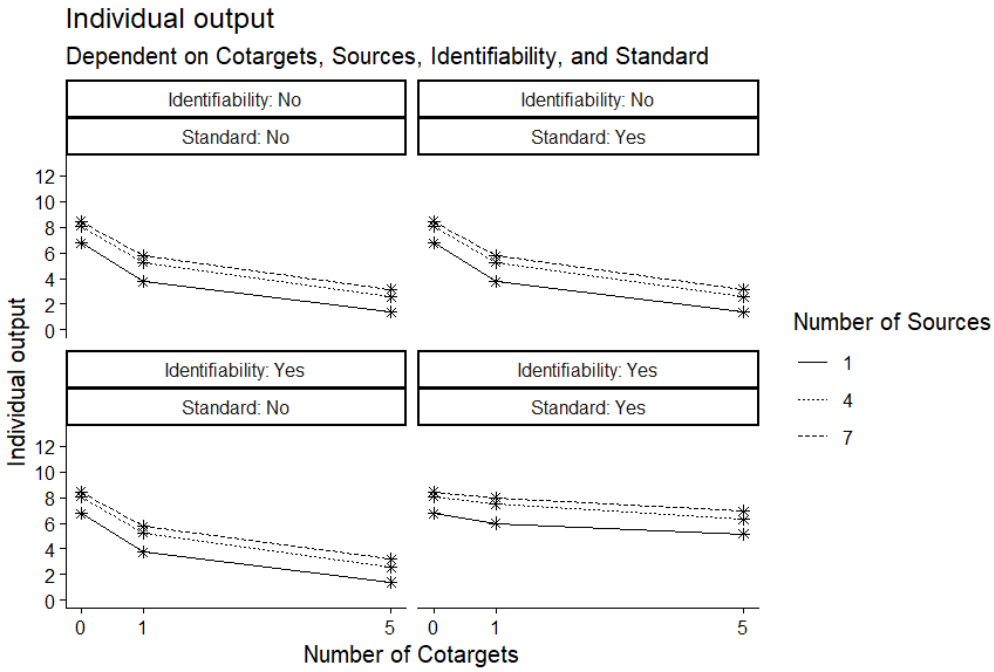


Figure 8

The superfunction without random noise



Evaluation of the Formal Model

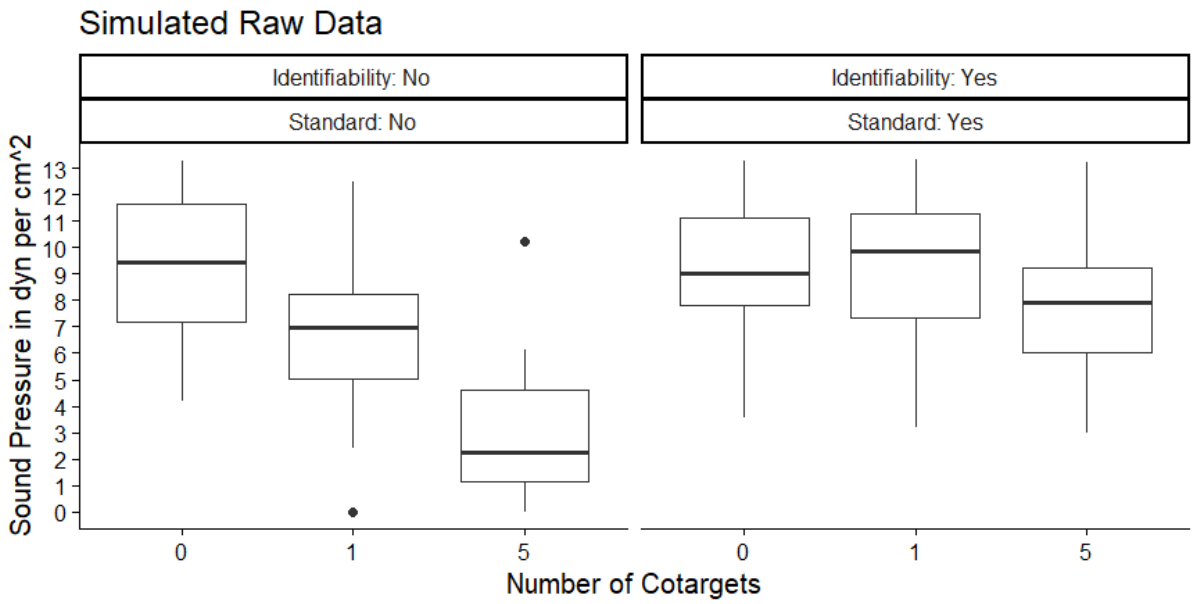
We wanted to replicate Latané et al. (1979) as well as possible by simulating n = 36 persons, each of whom had to participate in six experimental conditions: either zero, one, or five CT were present with two S each, once without ID and SA given and once with them. Note that the simulation with ID and SA (three experimental conditions) represents an extension and was not part of Latané et al. (1979).

MC had a predetermined expected value of 12.29 and was simulated using a beta distribution with $\alpha = 4.61$ and $\beta = 0.39$. IO, here the sound pressure in dyn/cm^2, was computed using the superfunction for IO, which includes random noise for each IE value. This random noise followed a normal distribution ($M = 0$, $SD = 0.2$).

The left panel shows the boxplots of the simulated raw data, which were examined in the first ANOVA, and the right panel the data for the second ANOVA. Measured was the individual sound output (IO) in dyn/cm^2. With ID and SA not given, the left panel shows the expected decrease in IO with growing group size. With ID and SA given (right panel), one cannot see this sharp decline, which is in line with the assumption of Harkins & Jackson (1985) (Figure 9).

Figure 9

Boxplots



We calculated an Analysis of Variance (ANOVA) like in Latané et al. (1979).

Because each ANOVA contained an alternative hypothesis with an "and"-link, there was no need for correcting the p-value. Due to the fact that the two ANOVAs were indirectly linked with an "or" (as they are independent and do not have to be both given), p-values were corrected for $n = 2$ ANOVAs with the help of the Bonferroni method (like an alternative hypothesis with an "or"-link) (OERLPMD, 2025a). Based on a significance level of $\alpha = .005$, p-values will be compared in this report with the significance level of $\alpha = .0025$.

We have conducted individual comparisons for "H_{01}: CT0-CT1≤0" and "H_{02}: CT1-CT5≤0". So, our simulated study would indicate that social loafing can be found in the population of our random sample and, therefore, be evidence that our model, which is strongly based on Latané et al. (1979), can produce the phenomenon if, for the first ANOVA, both p-values are significant. For the second ANOVA, rather the opposite can be stated (but one must be careful, because we only tested it for CT = 0, 1, and 5).

Regarding the first ANOVA with ID = 0 and SA = 0:

The result suggested that the mean individual output differs in at least two CT conditions in the population of our random sample ($F(2, 105) = 61.96$, $p < .0001$, $\eta^2 = .54$).

Both p-values were smaller than $\alpha = .0025$ (H_1: $p < .0001$, H_2: $p < .0001$), which is an indication for the linked alternative hypothesis, and we assume, as expected, that the individual output decreases with increasing group size (CT = 0, 1, and 5).

Regarding the second ANOVA with ID = 1 and SA = 1:

The result suggested that the mean individual output does not differ in at least two CT conditions in the population of our random sample ($F(2, 105) = 5.52$, $p = .0052$, $\eta^2 = .09$).
Only one p-value was smaller than $\alpha = .0025$ (H_1: $p = .5645$, H_2: $p = .0019$), which is an indication for the linked null hypothesis, and we assume, as expected, that the individual output does not decrease with increasing group size (CT = 0, 1, and 5).

Meta-Reflection

To reflect the process: The more we went in depth, the more we encountered obstacles - and as Socrates would say: knowing you know nothing (Cicero, 1997). E.g., during clarification of the definition and mechanism, we had to expand it, especially regarding CON and INC, since it was not described in much detail, and some small misunderstandings did occur due to the jingle-jangle problem present in Latané et al. (1979) (see Appendix C for VAST with a few jingle-jangle fallacies) (Kelley, 1927). Also, the functional relationships were very vaguely introduced.
Furthermore, Latané et al. (1979) repeatedly mentioned S in the context of social loafing but did not vary them, and we thought there might be the possibility of CT being S of P. But both of these issues are more the focus of the social facilitation research than the social loafing, the latter of which typically involves no evaluation of the performance of CT (Harkins et al., 1980; Karau & Williams, 1993). However, this evaluation potential was investigated subsequently for social loafing (Harkins & Jackson, 1985; Karau & Williams, 1993).
Overall, Latané et al. (1979) were sometimes a bit diffuse and imprecise.

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Appendix A: Construct and Variable Tables

Table A1

Construct Table

ID	**Type** - **P**henomenon - **C**oncept - **HOC**	**Short name**	**Quote**	**Reference**	**Rel. type (n, p, i, r, ...)**	**Comment (e.g., identified gaps & inconsistencies)**	**Incl. (Y/N)?**
CT	C	Co-Targets	"The more people who are the target of this pressure, [...]" (Latané et al., 1979, S.823) "According to Latane's (1981) social impact theory, people can be viewed as […] targets of social impact." (Karau & Williams, 1993, S. 682) AND	Latané et al. (1979) AND Karau & Williams (1993)	n1, p1, p2, n8		Y

			“and number of group members” (Latané et al., 1979, S. 823) “impact is divided among the group members” (Latané et al., 1979, S. 830) “However, if a person is a member of a group that is the target of social forces from outside the group” (Latané et al., 1979, S. 823)				
S	C	Source of pressure	“The impact that the experimenters have on an individual [...]” (Latané et al., 1979, S.825)	Latané et al. (1979) AND Karau & Williams (1993)	n2, p2	These persons are not described as a pressure precisely, but we can assume that they are the reason why the pressure occurs.	Y

			“According to Latane's (1981) social impact theory, people can be viewed as either sources […] of social impact.” (Karau & Williams, 1993, S. 682)			Pressure = Impact	
CON	C	Felt contingency between input and outcome	“[...] participants felt that the contingency between their input and the outcome [...]” (Latané et al., 1979, S. 830) “perceived contingency” (Karau & Williams, 1993, S. 684)	Latané et al. (1979) AND Karau & Williams (1993)	p1, n3, t1	Here, contingency is simply explained through the word itself, no further definition.	Y
INC	C	Incentive	“[...] incentive to work equally hard in groups of different	Latané et al. (1979)	t1, n4, p3		Y

			sizes.” (Latané et al., 1979, S. 830)				
P	C	impact/ pressure	“the amount of social pressure on the target person” (Latané et al., 1979, S. 823) “pressures on each individual” (Latané et al., 1979, S. 830) “In a group performance situation in which pressures to work come from outside the group” (Latané et al., 1979, S. 830) AND “[...] the impact of these forces	Latané et al. (1979)	p2, n5, n6, p3	Impact and pressure are not really distinctive concepts because impact can officiate as a social facilitation. Here, we assume they are the same because we limited our scope to the negative form of social impact - no facilitation at all.	Y

			on any given member [...]" (Latané et al., 1979, S. 823)				
IE	C	individual effort	"individual effort" (Latané et al., 1979, S. 822 ff.)	Latané et al. (1979)	p3, n7, t2	In this paper, there is no real distinction between "effort" and "output". These concepts were used partially synonymously. The problem is, they were not defined.	Y
Psy	HOC	Psyche	no citation	/	n11	We defined this HOC as the space in which the mental process is defined.	Y
IO	C	individual output	"in this case n times the average individual output" (Latané et al., 1979, S. 825)	Latané et al. (1979)	t2, n10, n12, n13		
			AND				

			"Since individual scores" (Latané et al., 1979, S. 830) AND "leading to an apparent drop in individual performance, a phenomenon we call social loafing" (Latané et al., 1979, S. 825) "sum of their individual performances" (Latané et al., 1979, S. 829)				
MC	C	individual maximum capacity	"dyads pulled at 91% of the sum of their individual capacities, trios at 82%, and groups of six at only 78%" (Latané	Latané et al. (1979)	n9, t2	Maximum capacity of an individual when infinite sources and 0 cotargets are present. (No citations are available; this is our own interpretation to	Y

			et al., 1979, S. 825)			simulate the phenomenon.)	
ID	C	Identifiability	”when scores were individually identifiable” (Latané et al., 1979, S. 830)	Latané et al. (1979)	n14, p1		Y
SA	C	Standard	”there must be a standard” (Karau & Williams, 1993, S. 683) “standard was available with which their outputs could be compared” (Karau & Williams, 1993, S. 688) “a standard against which this output can be compared” (Harkins & Jackson, 1985, S. 463)	Karau & Williams (1993) AND Harkins & Jackson (1985)	n15, p1		Y

Note. IDs, Types, Short Names, Quotes, References, Relationship types, Comments, and included in VAST for the Core Constructs of the Theory or rather the Mechanism.

Table A2

Variable Table

Construct	Source (if given)	Scale level	Range/Values	Anchors
CT	"and *number* of group members" (Latané et al., 1979, S. 823)	discrete	$\in \mathbb{N}_0$	0 = "zero cotargets are present" n = "number of cotargets (other persons in the evaluated group, number of group members minus 1)"
S	"*number* of these other persons." (Latané et al., 1979, S. 823) AND "Thus, as the number of performers decreased, the size of the audience increased. […] the larger the audience […]" (Latané et al., 1979, S. 826)	discrete	$\in \mathbb{N}_0$	0 = "zero persons that cause the social pressure" n = "number of persons that cause the social pressure"
P		continuous	$\in [0; 1[$	0 = "zero persons are the source of social pressure", "no social pressure" 1 = "maximal social pressure (a theoretical amount if infinite

				sources and zero cotargets are present)”
CON	“Individuals could "hide in the crowd" (Davis, 1969) and avoid the negative consequences of slacking off, or they may have felt "lost in the crowd" and unable to obtain their fair share of the positive consequences for working hard.” (Latané et. al, 1979, S. 830) AND “reward” (Latané et. al, 1979, S. 830)	continuous	∈ [0; 1]	0 = “expectation that the share (reward/punishment) of each member is completely independent of what they have given” 1 = “expectation that their share (reward/punishment) is equal to what they have given”
INC		continuous	∈ [0; 1]	0 = “no incentive to give any individual effort” 1 = “incentive to give full effort”
IE	e.g. “Thus the collective group performance, while increasing somewhat with group size, was substantially less than the sum of the	continuous	∈ [0; 1]	0 = “the person shows no effort at all to bring in some contribution to the group task” 1 = “the person shows their maximal effort in

	individual efforts, with dyads pulling at 93% of the sum of their individual efforts, trios at 85%, and groups of eight at only 49%." (Latané et al., 1979, S. 823)			contributing to the group task"
IO		continuous	∈ [0; inf[	depends on the task; in Latané et al. (1979): 0 = 0 dynes/cm^2 n = n dynes/cm^2
MC		continuous	∈ [0; inf[	depends on the task; in Latané et al. (1979): 0 = 0 dynes/cm^2 n = n dynes/cm^2
ID	"Since individual scores are unidentifiable" (Latané et al., 1979, S. 830) AND "This line of reasoning suggests that if inputs were made identifiable" (Latané et al., 1979, S. 830) AND	binary	∈ {0, 1}	0 = "individual scores are pooled" 1 = "individual scores are identifiable"

	"participants' outputs either were individually identifiable or were pooled" (Harkins & Jackson, 1985, S. 457)			
SA	"standard was available with which their outputs could be compared" (Karau & Williams, 1993, S. 688)	binary	∈ {0, 1}	0 = "a standard for comparison is not available" 1 = "a standard for comparison is available"

Note. Scale level, Range/Values, and Anchors for the core constructs of the theory. The citations are a basis for our Range/Values.

Appendix B: Relevant quotations for (functional) relationships

Z1: “It may be that participants felt that the contingency between their input and the outcome was lessened when performing in groups.” (Latané et al., 1979, S. 830)

Z2: “Only when performing alone can individual outputs be exactly evaluated and rewarded.” (Latané et al., 1979, S. 830)

Z3: “Thus, under these assumptions, if his own performance cannot be individually monitored, an individual's incentive to perform should be proportional to 1/n.“ (Latané et al., 1979, S. 830)

Z4: “This line of reasoning suggests that if inputs were made identifiable and rewards contingent on them, even when in groups, it would be impossible for performers to get a free ride and they would have an incentive to work equally hard in groups of different sizes.” (Latané et al., 1979, S. 830)

Z5: “We believe that social loafing occurs because there is usually a stronger perceived contingency between individual effort and valued outcomes when working individually. When working collectively, factors other than the individual's effort frequently determine performance, and valued outcomes are often divided among all of the group members.“ (Karau & Williams, 1993, S. 684)

Z6: “However, if a person is a member of a group that is the target of social forces from outside the group, the impact of these forces on any given member should diminish in inverse proportion to the strength, immediacy, and number of group members. Impact is divided up among the group members, in much the same way that responsibility for helping seems to be divided among witnesses to an emergency (Latane & Darley, 1970).” (Latané et al., 1979, S.823)

Z7: “The more people who are the target of this pressure, the less pressure should be felt by any one person.” (Latané et al., 1979, S. 823)

Z8: “The impact that the experimenters have on an individual seems to decrease as the number of coperformers increases, leading to an apparent drop in individual performance.” (Latané et al., 1979, S. 825)

Z9: “If a person is the target of social forces, increasing the number of other persons also in the target group should diminish the pressures on each individual because the impact is divided among the group members. In a group performance situation in which pressures to work come from outside the group and individual outputs are not identifiable, this division of impact should lead each individual to work less hard.” (Latané et al., 1979, S. 830)

Z10: “Social impact theory suggests that the experimenter's request to try as hard as possible on the task should be divided across targets, resulting in reduced effort as group size increases.” (Karau & Williams, 1993, S. 682)

Z11: “Social impact theory holds that when a person stands as a target of social forces coming from other persons, the amount of social pressure on the target person should increase as a multiplicative function of the strength, immediacy, and number of these other persons.” (Latané et al., 1979, S.823)

Z12: “the *n*th person should have less effect than the (*n* - l)th.” (Latané et al., 1979, S. 823)

Z13: “Just as perceptual judgments of physical stimuli follow power functions (Stevens, 1957), so also should judgments of social stimuli, and the exponent of the psychosocial power function should have an exponent of less than one, resulting in a marginally decreasing impact of additional people.“ (Latané et al., 1979, S. 830)

Z14: “Latané further suggests that just as psychophysical reactions to external stimuli can be described in terms of a power law (Stevens, 1957), so also should reactions to social stimuli, but with an exponent having an absolute value less than 1, so that the *n*th person should have less effect than the (*n* - l)th.“ (Latané et al., 1979, S. 823)

Z15: “Likewise, in our experiments, people may have felt a loss of control over their fair share of the rewards when they performed in groups, leading them also to become, if not lethargic and depressed, at least less enthusiastic about making lots of noise.

Since people were asked to shout both alone and in groups, they may have felt it smart to save their strength in groups and to shout as lustily as possible when scores were individually identifiable, marshalling their energy for the occasions when they could earn rewards. This line of reasoning suggests that if inputs were made identifiable and rewards contingent on them, even when in groups, it would be impossible for performers to get a free ride and they would have an incentive to work equally hard in groups of different sizes.“ (Latané et al., 1979, S. 830)

Z16: “The more people who are the target of this pressure, the less pressure should be felt by any one person. Since people are likely to work hard in proportion to the pressure they feel to do so, we should expect increased group size to result in reduced efforts on the part of individual group members.“ (Latané et al., 1979, S. 823)

Z17: “The impact that the experimenters have on an individual seems to decrease as the number of coperformers increases, leading to an apparent drop in individual performance” (Latané et al., 1979, S. 825)

Z18: “If a person is the target of social forces, increasing the number of other persons also in the target group should diminish the pressures on each individual because the impact is divided among the group members. In a group performance situation in which pressures to work come from outside the group and individual out puts are not identifiable, this division of impact should lead each individual to work less hard.” (Latané et al., 1979, S. 830)

Z19: “Social impact theory suggests that the experimenter's request to try as hard as possible on the task should be divided across targets, resulting in reduced effort as group size increases.” (Karau & Williams, 1993, S. 682)

Z20: “Thus, social impact theory suggests that the amount of effort expended on group tasks should decrease as an inverse power function of the number of people in the group.“ (Latané et al., 1979, S. 830)

Z21: “With respect to the Ringelmann phenomenon, social impact theory suggests that at least some of the effect should be due to reduced efforts on the part of group participants, and that

this reduced effort should follow the form of an inverse power function having an exponent with an absolute value less than one." (Latané et al., 1979, S. 823)

Z22: "These data suggest that to eliminate social loafing participants must feel not only that their outputs are individually identifiable […], but also that these outputs can be evaluated through comparison […]" (Harkins & Jackson, 1985, S. 457)

Z23: "This analysis suggests that evaluation potential plays a central role in […] loafing effects […]. We would suggest that two pieces of information must be known: the participant's individual output and a standard against which this output can be compared." (Harkins & Jackson, 1985, S. 463)

Appendix C: VAST with jingle-jangle fallacies

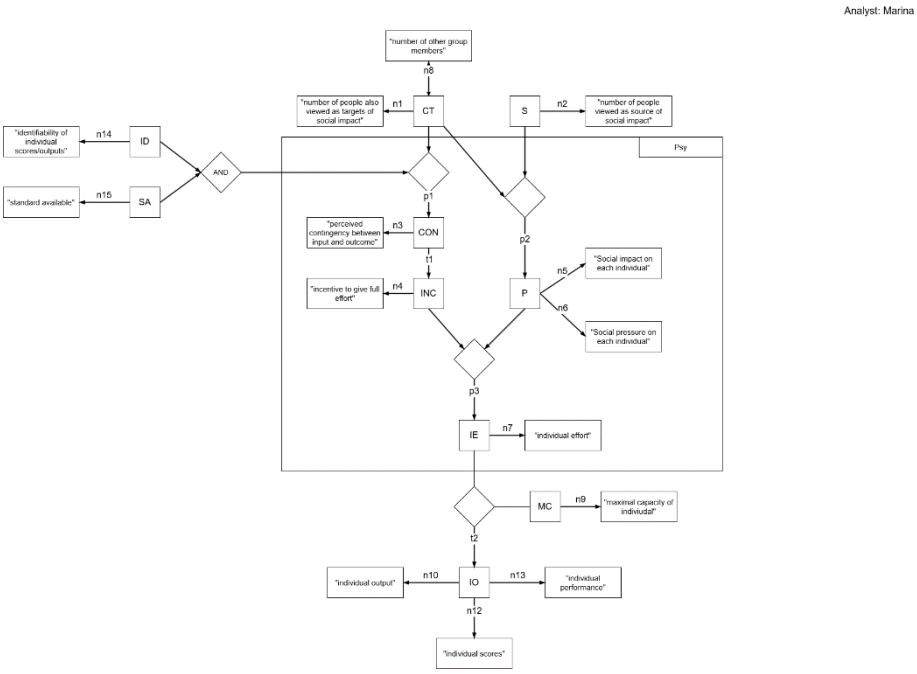

Figure C1. This is the same visual overview of the mechanism as in Figure 1, just with a few jingle-jangle fallacies.

Appendix D: Some statements that did not make it into the report due to shortenings

Identification of Relevant Phenomena

For the sake of completeness, though, there is the need to say that solely basing your decision on this homogeneity test to choose a random-effects model shouldn't be encouraged because this test can suffer from low power (Barili et al., 2018).

Regarding UTOS: There is the need to emphasize the fact that the majority were college students and from Western culture. The tendency was for easy, maximizing, additive, and physical or cognitive tasks. There was a larger percentage share of laboratory experiments.

Formulation of the (Proto-)Theory

One could argue that ID (and therefore SA) could also have an effect on P because evaluation potential also plays a role in facilitation effects (Harkins & Jackson, 1985), and due to common sense, one could thereby feel more pressure. But as it was only stated in the context of CON and INC, which is why I decided against it.

Development of a Formal Model

Regarding our scope: We limited our model to additive, maximizing tasks because Latané et al. (1979) used them in their study, even though social loafing was also found for, e.g., compensatory tasks (Karau & Williams, 1993). Additionally, it was shown that under the right conditions for disjunctive tasks, even if identifiability is given, there was a reduction in effort (Karau & Williams, 1993).

Regarding the beta distribution for MC: We based the mean value on the mean value in Latané et al. (1979), which we scaled up to 100% by using our established functions.

Regarding p1: One could argue that, first, if only one of the two variables (either ID or SA) is given, it could have a small impact on contingency. But it was argued that both of them have to be present for an effect (Harkins & Jackson, 1985; Karau & Williams, 1993), which was also empirically confirmed (Harkins & Jackson, 1985). And second, that noise would be appropriate in case, e.g., someone has a feeling they were rated worse because of sympathy - but to limit the scope (and also due to the noise already added to IE), this was not considered.

Regarding t1: Additionally, it could be plausible to suppose that INC does not equal CON because of other factors such as a task being intrinsically interesting or meaningful (Karau & Williams, 1993). To reduce the scope, these factors were not considered. But this was rather based on common sense.

Evaluation of the Formal Model

Regarding the normal distribution for the random noise: We had no value on which we could have based the *SD* on, and therefore we had to choose one ourselves.

Regarding the ANOVAs: Normally, you would take the normal distribution and variance equality assumptions into account (OERLPMD, 2025c).

For the first ANOVA: The estimated confidence intervals were CT0 - CT1: 95% CI [1.38, 3.72] and CT1 - CT5: 95% CI [2.79, 5.13].

For the second ANOVA: The estimated confidence intervals were CT0 - CT1: 95% CI [-1.35, 1.15] and CT1 - CT5: 95% CI [0.61, 3.11].

Eigenständigkeitserklärung

Hiermit erkläre ich, dass ich die vorliegende Arbeit selbstständig und ohne fremde Hilfe verfasst habe. Ich habe keine anderen als die angegebenen Quellen und Hilfsmittel benutzt und alle wörtlich oder sinngemäß übernommenen Stellen als solche kenntlich gemacht.

Pullhausen, 16.03.2026
Ort, Datum

Unterschrift

KI Statement

Statement on the use of generative AI and AI-assisted technologies in the writing process

In writing this paper, I have used the following tool/s: LanguageTool, DeepL, ChatGPT. I used it for the following purpose/s:

- ☐ improving linguistic quality and readability
- ☐ writing assistance e.g. for generating table of contents, outlines, first sentences and paragraphs
- ☐ finding relevant citations (correctly marked as such)
- ☐ finding relevant scientific sources (correctly marked as such)
- ☐ translation of citations or text passages (correctly marked as such, e.g. "translated with DeepL")
- ☒ summarizing information
- ☐ transcribing assistance from audio or video
- ☒ generating or optimizing programming code (e.g. R- or Python-Code)
- ☐ researching terms
- ☐ generating term definitions
- ☒ other [please specify]:
    - LanguageTool: Helping with punctuation, spelling, and grammar
    - DeepL: Translation of single words
    - ChatGPT: Asking if what I was thinking makes sense

After using this tool or service, I reviewed and edited the content as needed and I take full responsibility for the content of this paper. I certify that this paper does not contain lengthy passages (e.g. summary/abstract, or full paragraphs) of purely AI-generated text.

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